



1. INTRODUCTION



1.1 Problem

Many people exercise without professional supervision.



1.2 Motivation

To develop an intelligent, non-intrusive, and cost-effective system that uses AI and computer vision to analyse exercise posture in real-time and provide instant feedback.



1.3 Objectives

- 1 Analyze body posture during exercise.
- 2 Calculate joint angles accurately.
- 3 Provide real-time feedback to correct posture.

2. METHODOLOGY

Planning & Analysis

Data Collection & Analysis

System Design

</> Implementation & Development

1

Person Detection (YOLOv5)

(👤)

2

Pose Estimation (MediaPipe)

(👤)

3

Joint Angle Calculation + Random Forest & MLP Classification

(📐)

4

Visual Feedback + LLM

(💬)

Testing & Evaluation

Unit Testing

Integration Testing

Live System Testing (Real Users)

Deployment & Refinement (Streamlit)

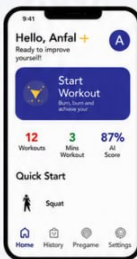
3. EXPERIMENTS & RESULTS

3.2 UI Output (Screens)

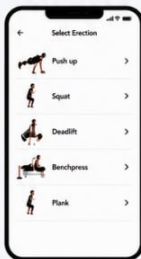
1 Startup Screen



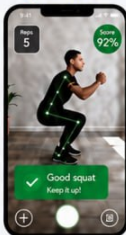
2 Main Page



3 Exercise Selection



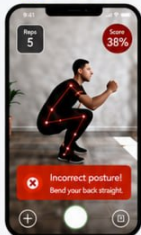
4 Correct Squat



5 Incorrect Workout



6 Incorrect Squat



3.3 FUTURE WORK

- Expand exercise library to include Yoga and Pilates.
- Integrate real-time voice guidance and audio feedback.
- Develop mobile apps for iOS and Android.
- Implement personalized workout plans and progress tracking.

4. CONCLUSION

The Coach Ai system provides an effective, real-time solution for exercise posture correction using computer vision. By eliminating the need for wearable sensors and providing immediate visual feedback, it empowers users to exercise safely and effectively at home. The system demonstrates high accuracy in posture detection, making it a valuable tool for fitness and rehabilitation.

5. REFERENCES

1. R. Bahr and I. A. Bahr, "Incidence of acute volleyball injuries: a prospective cohort study of injury mechanisms and risk factors." Scand. J. Med. Sci. Sports., vol. 7, no. 3, pp. 166-171, 1997.
2. B. Davis and A. Thompson, "Benchmarking Real-Time AI Systems: Metrics for Posture Correction Applications," IEEE Trans. Instrum. Meas., vol. 72, 2022.



TEAM MEMBERS

Anfal Alharbi 431201470
Hanady Alharbi 431201695
Eman Alghamari 431201312
Ghadeer almouazini 422025786



SUPERVISOR

Dr. Shabana Habib



COLLEGE

College of Computer
Information Technology